$t\bar{t}$ and $hh \to b\bar{b}\,\ell\nu\,q\bar{q}'$: samples, the ten angles, and $f_{t\bar{t}}$ by OSDE

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Outline

1. Introduction
2. $t\bar{t}$ samples and validation
3. The ten angles
4. hh samples
5. $f_{t\bar{t}}$ by OSDE
6. How many coefficients matter?
7. Acceptance
8. Summary

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Goal

- Generate $pp \to t\bar{t} \to b\bar{b}\,W^+W^- \to b\bar{b}\,\ell\nu\,q\bar{q}'$ at parton, particle (gen) and detector (reco) level
- Validate the samples and the ten angles $\mathbf{x} = \{Y, \Theta, \boldsymbol{\theta}, \tilde{\boldsymbol{\theta}}\}$ used for the $t\bar{t}$ density $f_{t\bar{t}}(\mathbf{x})$
- Everything is scripted: `generate_samples.py`, `make_plots.py`, `make_slides.py`

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Sample generation

- MadGraph5_aMC@NLO 3.6.7 (LO, `sm`, 4-flavour proton) → Pythia8 → Delphes 3.5.1, ATLAS card, **no pileup**
- Decay chains keep spin correlations; no generator-level cuts:

```
generate    p p > t t~, (t > b w+, w+ > lep+ nul), (t~ > b~ w-, w- > q q)
add process p p > t t~, (t > b w+, w+ > q q), (t~ > b~ w-, w- > lep- nul~)
```

- $\sqrt{s} = 13$ TeV, $\ell = e, \mu$, 10000 events, seed 12345
- $\sigma = 136.5$ pb $\approx \sigma^{\mathrm{LO}}_{t\bar{t}} \times 2\,\mathcal{B}(W \to \ell\nu)\,\mathcal{B}(W \to q\bar{q}')$ (with $\ell = e, \mu$)

Object definitions

	particle level (gen)	reco (Delphes ATLAS)
jets	GenJet anti-k_t 0.4	Jet anti-k_t 0.4
	$p_T > 25$ GeV, $\lvert\eta\rvert < 2.5$	
b-jets	$\Delta R(j,b) < 0.4$, $p_T^b > 5$ GeV	BTag bit 0
leptons	status-1 e/μ from W (τ)	Electron, Muon
	$p_T > 25$ GeV, $\lvert\eta\rvert < 2.47$ (e), 2.5 (μ)	
overlap	drop jets $\Delta R(j,\ell) < 0.4$	UniqueObjectFinder
E_T^{miss}	GenMissingET	MissingET

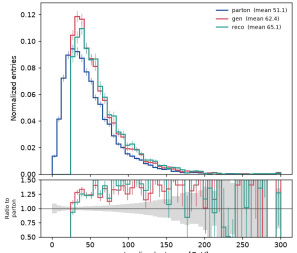
Preselection: exactly 1 lepton, ≥ 4 jets, ≥ 1 b-jet.

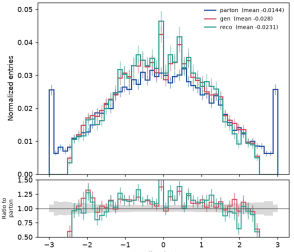
Cutflow

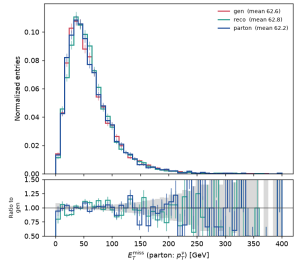
selection	gen		reco	
All events	10000	100.0%	10000	100.0%
exactly 1 lepton	7071	70.7%	4907	49.1%
≥ 4 jets	4196	42.0%	2830	28.3%
≥ 1 b-jet	4161	41.6%	2454	24.5%
≥ 2 b-jets	3305	33.0%	1221	12.2%
χ^2 reconstruction (from ≥ 1 b)	4134	41.3%	2440	24.4%

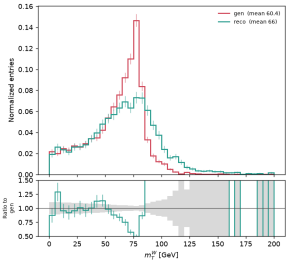
Fractions relative to all generated events.

Leptons and E_T^{miss}

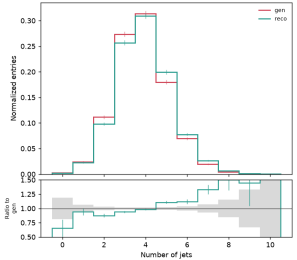


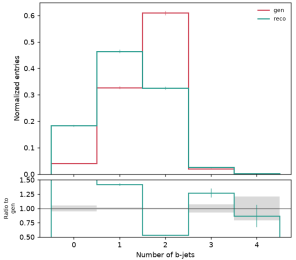


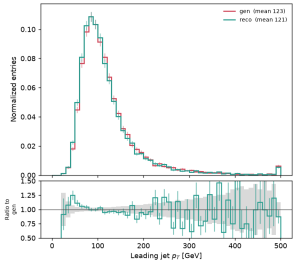


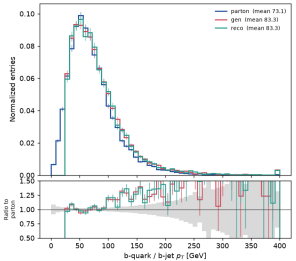


Jets


gen
reco
Normalised entries
Ratio to gen
Number of jets

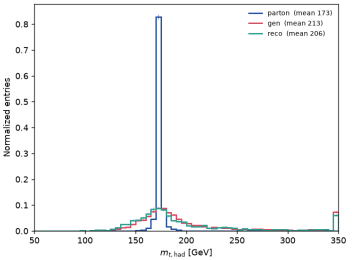

gen
reco
Normalised entries
Ratio to gen
Number of b-jets

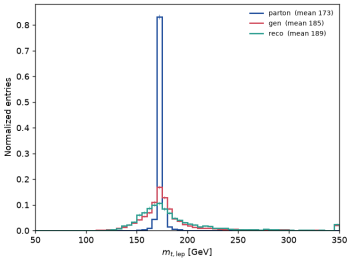

gen (mean 123)
reco (mean 121)
Normalised entries
Ratio to gen
Leading jet p_T [GeV]

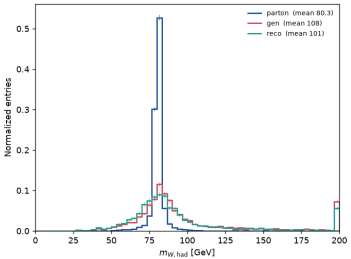

parton (mean 73.1)
gen (mean 83.3)
reco (mean 83.3)
Normalised entries
Ratio to parton
b-quark / b-jet p_T [GeV]

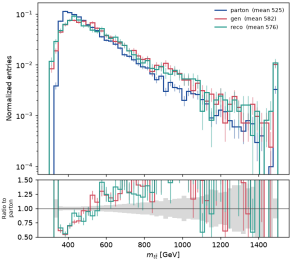
- $\chi^2 = \left(\frac{m_{jj}-m_W}{\sigma_W}\right)^2 + \left(\frac{m_{jjb}-m_t}{\sigma_t}\right)^2 + \left(\frac{m_{\ell\nu b}-m_t}{\sigma_t}\right)^2$ over assignments of the 6 leading jets
- b slots hold $\min(n_b, 2)$ b-tagged jets
- p_z^ν from the m_W constraint; both solutions enter the χ^2, so the m_t term resolves the two-fold ambiguity
- All four jets correct in 23% (gen) and 18% (reco) of reconstructed events

Reconstructed masses









Mass resolution (median $\pm$ 68% half-width, GeV)

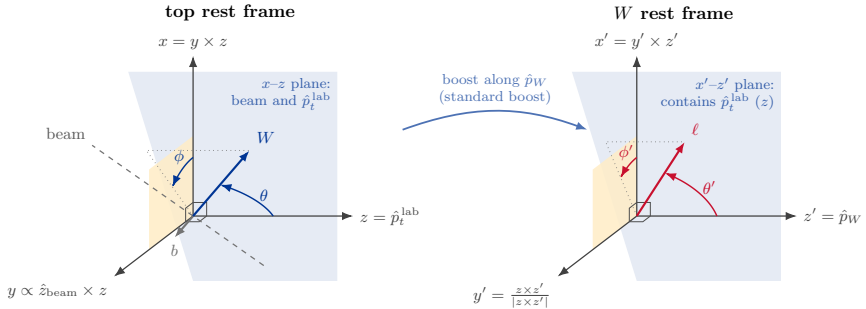
	$m_{t,\mathrm{had}}$, all	$m_{t,\mathrm{had}}$, correct jets	$m_{W,\mathrm{had}}$, all	$m_{W,\mathrm{had}}$, correct jets
gen	182 ± 47	175 ± 14	85 ± 29	83 ± 9
reco	177 ± 44	170 ± 16	84 ± 25	80 ± 10

The broad inclusive peaks come from wrong jet choices (combinatorics) and quarks that are out of acceptance, not from the jet response.

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Definition of the ten angles

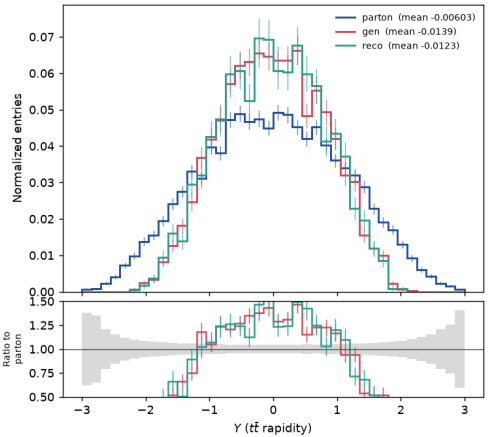


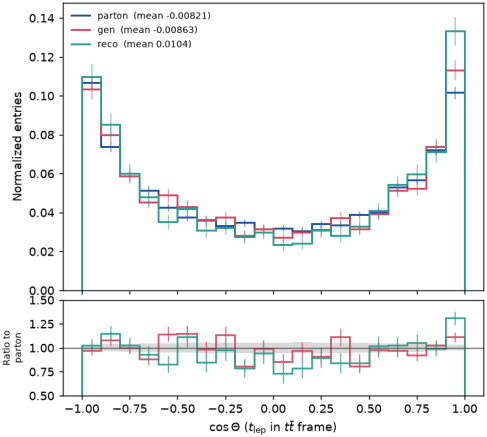
Top rest frame: $z = \hat{p}_t^{\text{lab}}$, $y \propto \hat{z}_{\text{beam}} \times z$ (normal to the plane of the beam and the top), $x = y \times z$; (θ, ϕ) is the direction of the W, ϕ measured from x in the xy plane (gold).
W **rest frame**: $z' = \hat{p}_W$, $y' = z \times z'/|z \times z'|$ lies in the xy plane, $x' = y' \times z'$; (θ', ϕ') is the direction of the charged lepton, i.e. $(x', y', z') = R(\phi, \theta, 0)\,(x, y, z)$.
Hadronic top: the same construction gives $(\tilde{\theta}, \tilde{\phi}, \tilde{\theta}', \tilde{\phi}')$, with the down-type quark in place of ℓ.

The ten angles (as in the paper)

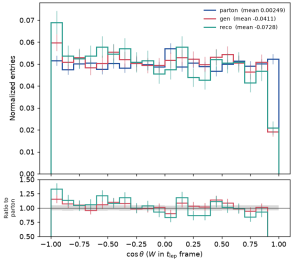
- Y: $t\bar{t}$ rapidity (lab); Θ: polar angle of t_{lep} in the $t\bar{t}$ frame
- (θ, ϕ): W in the top rest frame; $z = \hat{p}_t^{\text{lab}}$, $y = \hat{z}_{\text{beam}} \times z$, $x = y \times z$
- (θ', ϕ'): fermion in the W rest frame (boost from top frame); $z' = \hat{p}_W$, $y' = z \times z'$, $x' = y' \times z'$, i.e. $D(\phi, \theta, 0)$
- Leptonic top: fermion $= \ell$. Hadronic top ($\sim$): fermion = down-type quark
- gen/reco: down-type jet chosen by truth-matching to the LHE quarks (assumes up/down tagging, as in the paper)
- Check: building momenta from known angles and recovering them agrees to 10^{-14}

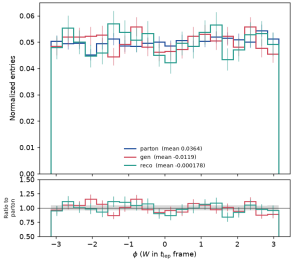
Production angles Y, $\cos\Theta$

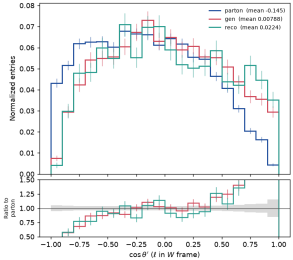


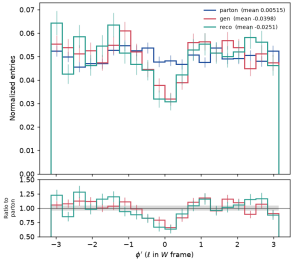


Leptonic top: $\theta, \phi, \theta', \phi'$

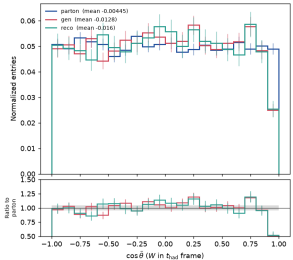


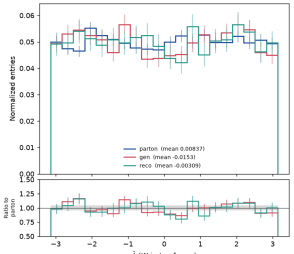


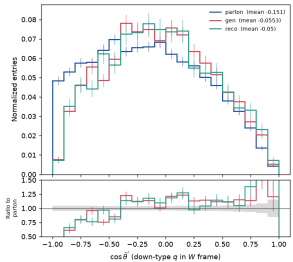


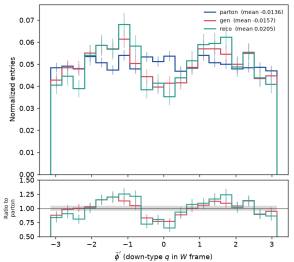


Hadronic top: $\tilde{\theta}, \tilde{\phi}, \tilde{\theta}', \tilde{\phi}'$









W helicity fractions from $\cos\theta'$ (moments)

	F_0	F_L	F_R
parton, ℓ	0.715 ± 0.013	0.288 ± 0.008	-0.003 ± 0.008
parton, down-type q	0.705 ± 0.013	0.298 ± 0.008	-0.003 ± 0.008
gen, ℓ	0.769 ± 0.019	0.107 ± 0.012	0.123 ± 0.012
gen, down-type q	0.954 ± 0.017	0.078 ± 0.011	-0.032 ± 0.011
reco, ℓ	0.687 ± 0.026	0.134 ± 0.017	0.179 ± 0.017
reco, down-type q	0.929 ± 0.023	0.086 ± 0.015	-0.014 ± 0.015

SM (LO): $F_0 = m_t^2/(m_t^2 + 2m_W^2) \approx 0.70$, $F_L \approx 0.30$, $F_R \approx 0$. Parton level reproduces it; gen/reco are distorted by acceptance and combinatorics (no unfolding).

Angle resolution vs parton level (correlation / median $|\Delta|$)

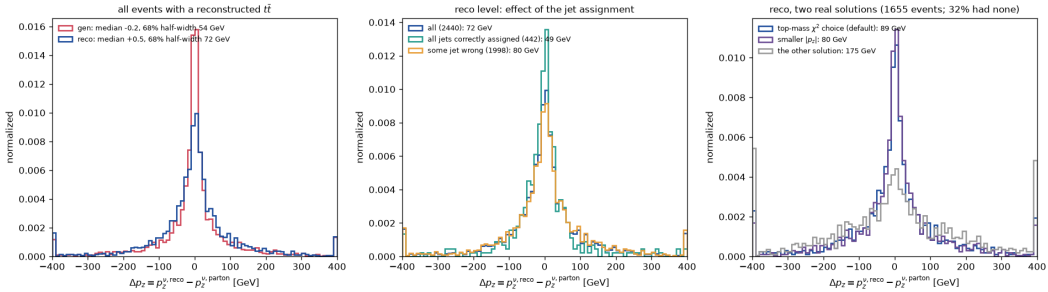
angle	gen, all	gen, correct jets	reco, all	reco, correct jets
Y	0.90 / 0.16	0.99 / 0.05	0.89 / 0.18	0.98 / 0.06
$\cos\Theta$	0.67 / 0.13	0.97 / 0.04	0.64 / 0.15	0.97 / 0.06
$\cos\theta$	0.70 / 0.14	0.89 / 0.06	0.62 / 0.22	0.84 / 0.11
ϕ	0.48 / 0.36	0.82 / 0.12	0.40 / 0.52	0.89 / 0.16
$\cos\theta'$	0.68 / 0.11	0.92 / 0.04	0.60 / 0.20	0.89 / 0.11
ϕ'	0.45 / 0.37	0.68 / 0.13	0.30 / 0.63	0.47 / 0.28
$\cos\tilde{\theta}$	0.59 / 0.18	0.98 / 0.04	0.45 / 0.24	0.97 / 0.05
$\tilde{\phi}$	0.40 / 0.46	0.88 / 0.09	0.34 / 0.69	0.90 / 0.09
$\cos\tilde{\theta}'$	0.41 / 0.18	0.97 / 0.04	0.31 / 0.25	0.96 / 0.06
$\tilde{\phi}'$	0.39 / 0.55	0.93 / 0.06	0.33 / 0.73	0.94 / 0.07

Neutrino p_z resolution

	median Δp_z [GeV]	68% half-width [GeV]
gen	-0.25	54
reco	0.5	72.2

$\Delta p_z = p_z^{\nu,\mathrm{reco}} - p_z^{\nu,\mathrm{parton}}$, events with a reconstructed $t\bar{t}$ system.

Neutrino Δp_z



Correct jets: 49 GeV vs 72 GeV for all events (68% half-width). No real solution in 32% of events. With two solutions, the top-mass χ^2 picks the one closer to the truth in 63% of events vs 65% for smaller $|p_z|$.

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- MadGraph5 built-in `heft` model (LO, $m_t \to \infty$), same Pythia8 + Delphes ATLAS chain

```
generate    g g > h h, (h > b b~), (h > lep+ nul q q)
add process g g > h h, (h > b b~), (h > lep- nul~ q q)
```

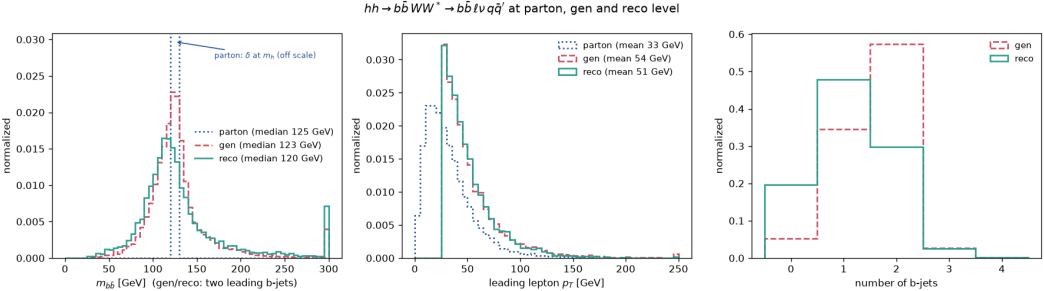
- $h \to WW^*$ has one off-shell W, so $h \to \ell\nu q\bar{q}'$ is a 4-body decay
- 20000 events, seed 2001; $\sigma \times \mathcal{B} = 0.112$ fb (LO HEFT with MG defaults; *not* an SM prediction, so plots use unit area)
- For tt-hypothesis quantities, hh uses the same χ^2; at parton level the b pairing is the one closest to m_t
- Prompt leptons: status-1 e/μ from W, Z, h, or a τ from one of these (in $h \to WW^*$ the lepton can come directly from the h)

Cutflow: $t\bar{t}$ vs hh (reco)

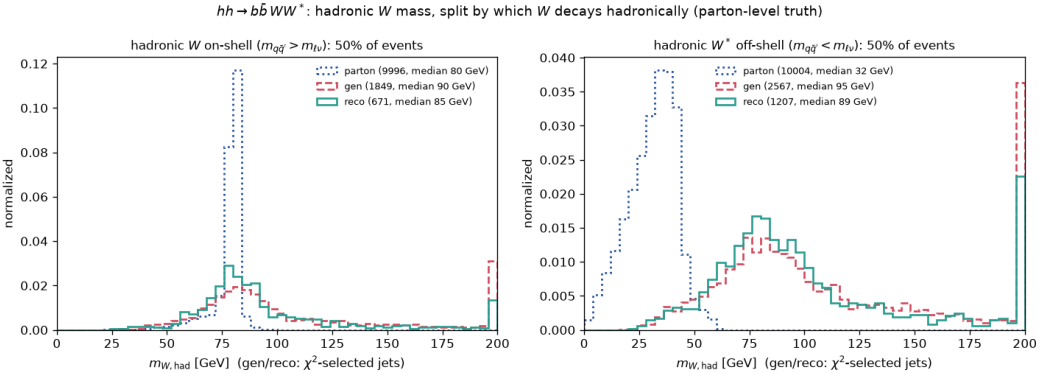
selection	$t\bar{t}$		hh	
All events	10000	100.0%	20000	100.0%
exactly 1 lepton	4907	49.1%	5403	27.0%
≥ 4 jets	2830	28.3%	2133	10.7%
≥ 1 b-jet	2454	24.5%	1887	9.4%
≥ 2 b-jets	1221	12.2%	920	4.6%
χ^2 reconstruction (from ≥ 1 b)	2440	24.4%	1878	9.4%

hh leptons are softer (from W^*) and less isolated, so the efficiency is lower.

hh sample: parton, gen and reco level



hh: hadronic W mass, on-shell W vs W^*



Hadronic boson is the W^* in 50% of events (true median 32 GeV), but the $t\bar{t}$-hypothesis χ^2 picks jet pairs near m_W (median 95 / 89 GeV at gen / reco): the reconstruction makes hh look like $t\bar{t}$.

Outline

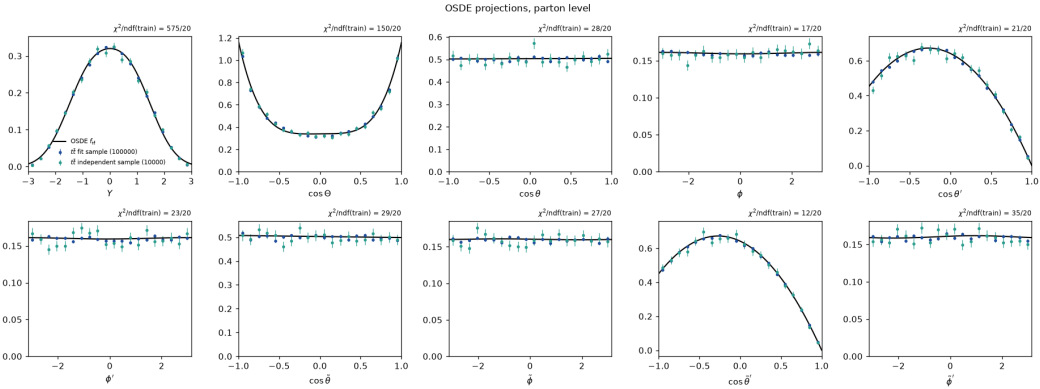
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Basis functions (as in the paper)

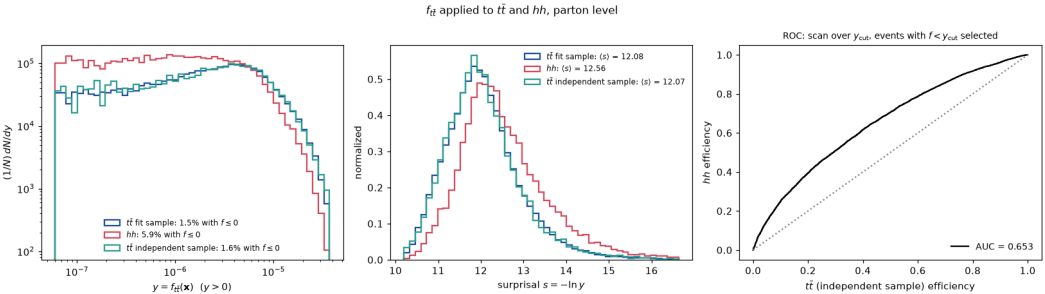
- $M^{j_1 j_2}_{m' m}(\phi, \theta, \phi', \theta') = \frac{\sqrt{2j_1+1}\sqrt{2j_2+1}}{4\pi} D^{j_1}_{m' m}(\phi, \theta, 0)\, D^{j_2}_{m 0}(\phi', \theta', 0)$, $j_1 \leq 1$, $j_2 \leq 2$: 24 per top
- $p_l(\cos\Theta) = \sqrt{\frac{2l+1}{2}} P_l$, $h_n(Y) = (2^n n! \sqrt{\pi})^{-1/2} H_n(Y) e^{-Y^2/2}$, $l < 5$, $n < 5$
- $B = M_{\mathbf{j}} M_{\tilde{\mathbf{j}}}\, p_l\, h_n$: $24 \times 24 \times 5 \times 5 = 14400$ coefficients
- OSDE: $\hat{c} = \frac{1}{N} \sum_k B^*(\mathbf{x}_k)$, with standard error; $f_{t\bar{t}} = \sum \hat{c}\, B$ (real, since the set is closed under conjugation)
- Hermite *functions*: bare H_n are orthogonal only with weight e^{-Y^2}
- **Three separate fits**: $f^{\text{parton}}_{t\bar{t}}$ from parton-level $t\bar{t}$ angles, applied to parton-level $t\bar{t}$ and hh; likewise f^{gen} (gen on gen) and f^{reco} (reco on reco)

Checks of the basis and of OSDE

- Unit tests: 7 passed
- Wigner $d^j_{m'm}$ agrees with closed forms and with `sympy` to 10^{-12}
- Gram matrices of M (exact quadrature), p_l, h_n equal the identity to 10^{-12}
- Toy: 2×10^5 events from a density with known coefficients; all 14400 recovered with pulls of unit width; projections and $\int f = 1$ reproduced
- Fitted f: imaginary part $< 10^{-10}$ of the real part

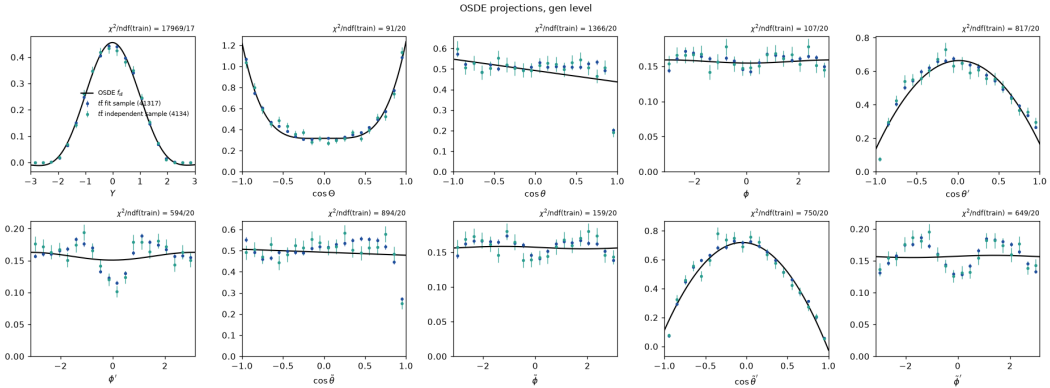


Black: $f_{t\bar{t}}$ integrated over the other nine angles. Largest χ^2/ndf vs training data: Y (575/20).



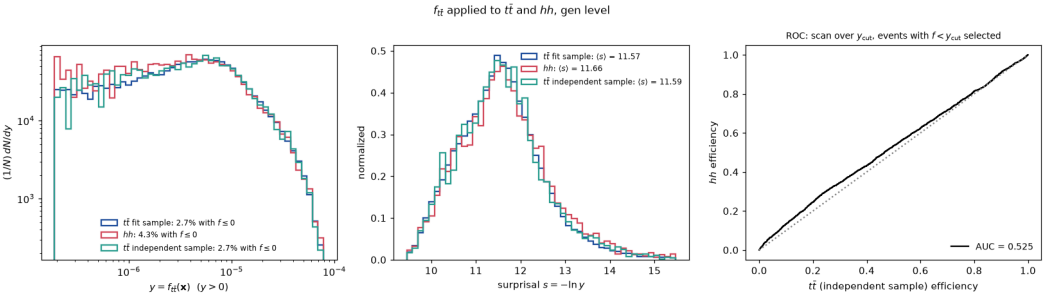
Left: $y = f_{t\bar{t}}(\mathbf{x})$; middle: surprisal $s = -\ln y$; right: hh vs $t\bar{t}$ efficiency when calling $f < y_{\text{cut}}$ anomalous.

$f_{t\bar{t}}^{\text{gen}}$ (fit on gen-level $t\bar{t}$, 41317 events): projections



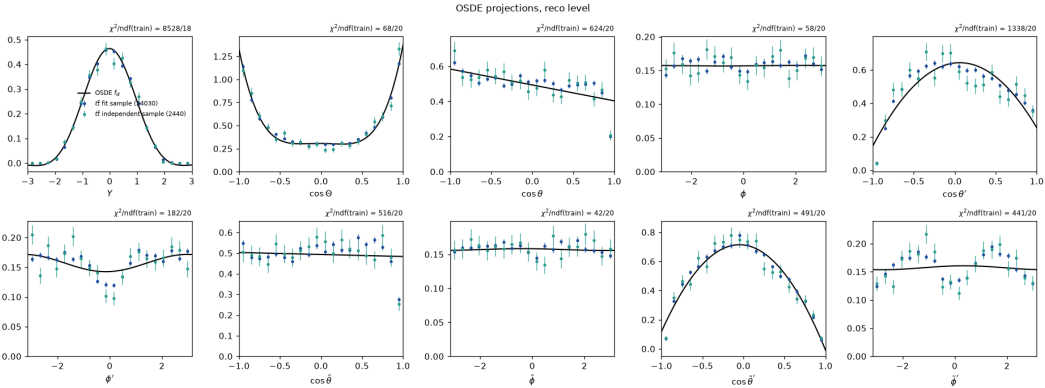
Black: $f_{t\bar{t}}$ integrated over the other nine angles. Largest χ^2/ndf vs training data: Y (17969/17).

$f_{t\bar{t}}^{\text{gen}}$ applied to gen-level $t\bar{t}$ (independent) and hh



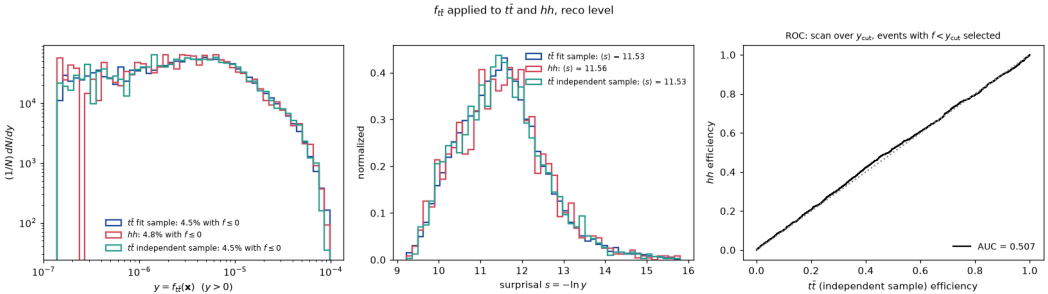
Left: $y = f_{t\bar{t}}(\mathbf{x})$; middle: surprisal $s = -\ln y$; right: hh vs $t\bar{t}$ efficiency when calling $f < y_{\text{cut}}$ anomalous.

$f_{t\bar{t}}^{\text{reco}}$ (fit on reco-level $t\bar{t}$, 24030 events): projections



Black: $f_{t\bar{t}}$ integrated over the other nine angles. Largest χ^2/ndf vs training data: Y (8528/18).

$f_{t\bar{t}}^{\text{reco}}$ applied to reco-level $t\bar{t}$ (independent) and hh



Left: $y = f_{t\bar{t}}(\mathbf{x})$; middle: surprisal $s = -\ln y$; right: hh vs $t\bar{t}$ efficiency when calling $f < y_{\text{cut}}$ anomalous.

$f_{t\bar{t}}$ summary

level	N_{train}	$\lvert\hat{c}\rvert > 3\sigma$	$\int f$	$f \leq 0$: $t\bar{t}/hh$	$\langle s \rangle$: $t\bar{t}/hh$	AUC
parton	100000	186	1.006	1.6% / 5.9%	12.07 / 12.56	0.653
gen	41317	330	0.985	2.7% / 4.3%	11.59 / 11.66	0.525
reco	24030	253	0.987	4.5% / 4.8%	11.53 / 11.56	0.507

$t\bar{t}$ numbers are from the independent test sample. $f \leq 0$ comes from the truncated series (Gibbs overshoot), as noted in the paper. AUC = probability that an hh event has lower $f_{t\bar{t}}$ than a $t\bar{t}$ event.

Fit sample vs independent sample: self-inclusion bias

level	self-term / f	$f \leq 0$: fit / fit LOO / indep.	$\langle s \rangle$: fit / fit LOO / indep.	KS vs indep.: fit / LOO
parton	7%	0.6% / 1.5% / 1.6%	11.98 / 12.08 / 12.07	0.041 / 0.010
gen	10%	1.0% / 2.7% / 2.7%	11.44 / 11.57 / 11.59	0.071 / 0.012
reco	16%	1.2% / 4.5% / 4.5%	11.33 / 11.53 / 11.53	0.103 / 0.012

Each fit event adds $\frac{1}{N}\sum_j |B_j(\mathbf{x}_k)|^2$ to its own f. Removing it (leave-one-out, LOO) makes the fit sample agree with the independent $t\bar{t}$ sample: the difference was entirely this bias, largest at reco (smallest N). Plots use LOO for the fit sample.

Diagnostics: statistical noise of the 14400 coefficients

level	all (paper)	$\lvert\hat{c}\rvert > 2\sigma$	$\lvert\hat{c}\rvert > 3\sigma$	$\lvert\hat{c}\rvert > 5\sigma$
parton	14400 / 0.653 / $+0.089$	751 / 0.663 / $+0.014$	186 / 0.666 / $+0.001$	54 / 0.669 / -0.005
gen	14400 / 0.525 / $+0.144$	827 / 0.525 / $+0.042$	330 / 0.525 / $+0.025$	153 / 0.527 / $+0.009$
reco	14400 / 0.507 / $+0.202$	712 / 0.507 / $+0.040$	253 / 0.504 / $+0.025$	127 / 0.507 / $+0.015$

Entries: coefficients kept / AUC(hh vs $t\bar{t}$ test) / $\langle s \rangle_{\text{test}} - \langle s \rangle_{\text{train}}$ (overfitting). Dropping insignificant terms removes the overfitting, but the gen/reco AUC stays ≈ 0.5, so the loss of separation is not caused by noise.

Diagnostics: Hermite scale for Y (χ^2 of the Y projection, train)

level	$s = 1.0$	$s = 0.9$	$s = 0.8$	$s = 0.7$	$s = \mathrm{std}(Y)$
parton	660/24	998/24	379/24	507/24	230/24
gen	2298/20	2956/20	548/20	728/20	548/20
reco	4485/21	3864/21	721/21	361/21	515/21

The paper uses $s = 1$ (default here); `osde_fit.py --y-scale` changes it. Gen/reco Y is narrower (std ≈ 0.8), so $s = 1$ fits poorly there and f overshoots below zero at large $|Y|$.

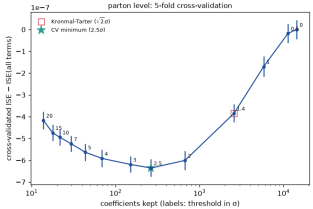
Outline

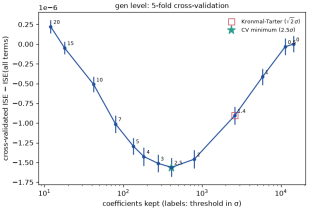
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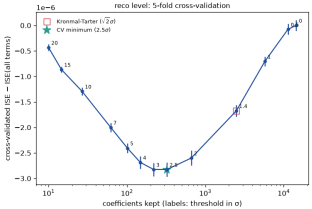
Criterion for dropping coefficients

- Keep a term iff $|\hat{c}|/\sigma(\hat{c}) > t$ (constant always kept). Conjugate partners have equal $|\hat{c}|$ and σ, so they are kept together and f stays real (checked)
- Kronmal–Tarter ($t = \sqrt{2}$): dropping a term costs $|c|^2$, keeping it costs σ^2; keep iff $|\hat{c}|^2 > 2\sigma^2$ (minimizes the integrated squared error per term)
- With 14400 mostly-null terms, $\sqrt{2}\sigma$ still admits $\sim$16% of the pure-noise terms
- Choose t by 5-fold cross-validation: for an orthonormal basis, $\mathrm{ISE} - \int f^2 = \sum_{\text{kept}} |\hat{c}|^2 - 2\,\mathrm{Re} \sum_{\text{kept}} \hat{c}\, \hat{c}^*_{\text{held-out}}$ (no integrals needed)
- CV minimum at $t = 2.5$: 326 (parton), 459 (gen), 342 (reco) of 14400 terms; flat out to 5σ: 54, 153, 127

Cross-validated error vs number of coefficients







Simplified $f_{t\bar{t}}$: performance

level	terms kept	number	AUC	$f \leq 0$ ($t\bar{t}$ indep.)	$f \leq 0$ (hh)	$\langle s \rangle$ fit LOO / indep.
parton	all (paper)	14400	0.653	1.6%	5.9%	12.08 / 12.07
	$\sqrt{2}\sigma$ (K–T)	2629	0.658	1.0%	4.8%	12.04 / 12.06
	CV best	326	0.665	0.2%	2.4%	12.06 / 12.06
	5σ	54	0.669	0.1%	2.1%	12.07 / 12.06
gen	all (paper)	14400	0.525	2.7%	4.3%	11.57 / 11.59
	$\sqrt{2}\sigma$ (K–T)	2725	0.526	1.9%	2.9%	11.53 / 11.57
	CV best	459	0.525	0.9%	1.3%	11.56 / 11.58
	5σ	153	0.527	1.0%	1.5%	11.57 / 11.58
reco	all (paper)	14400	0.507	4.5%	4.8%	11.53 / 11.53
	$\sqrt{2}\sigma$ (K–T)	2551	0.508	2.7%	2.8%	11.47 / 11.52
	CV best	342	0.505	1.1%	1.5%	11.52 / 11.54
	5σ	127	0.507	1.5%	1.9%	11.54 / 11.55

Dropping insignificant terms cuts the negative-f (Gibbs) fraction and keeps the hh separation: the simplified f is as good, with $\sim$1–3% of the terms.

Leading terms at parton level (after the constant)

#	term	$\lvert\hat{c}\rvert/\sigma$	physics
1	$P_2(\cos\Theta)$	115	production angle
2	$M_{0,0}^{01}$	88	W helicity (lep)
3	$\tilde{M}_{0,0}^{01}$	87	W helicity (had)
4	$M_{0,0}^{02}$	80	W helicity (lep)
5	$\tilde{M}_{0,0}^{02}$	79	W helicity (had)
6	$h_2(Y)$	59	rapidity
7	$P_4(\cos\Theta)$	34	production angle
8	$M_{0,0}^{01} \times P_2(\cos\Theta)$	32	W helicity (lep); production angle

M: leptonic top, $\tilde{M}$: hadronic top; $j_1 = 0$: W helicity, $j_1 = 1$: top spin. Production (P_l), W helicity and Y dominate.

What the significant ($> 5\sigma$) terms describe

physics content of the term	parton	gen	reco
production angle + top pol. $\times W$ hel. interference	–	30	26
rapidity + top pol. $\times W$ hel. interference	–	24	12
top pol. $\times W$ hel. interference + $t\bar{t}$ spin correlation	16	–	–
W helicity + production angle	12	8	6
production angle + rapidity + top polarization	–	12	12
production angle + rapidity + top pol. $\times W$ hel. interference	–	11	11
W helicity + production angle + rapidity	1	11	11
W helicity	8	6	6
top pol. $\times W$ hel. interference	–	7	7
production angle + rapidity	2	7	6
total	**53**	**152**	**126**

Parton: production, W helicity and $t\bar{t}$ spin correlations; imaginary parts consistent with 0 (max 1.4σ), as expected without CP violation. Reco: spin correlations no longer resolved; acceptance-induced "polarization" terms appear instead.

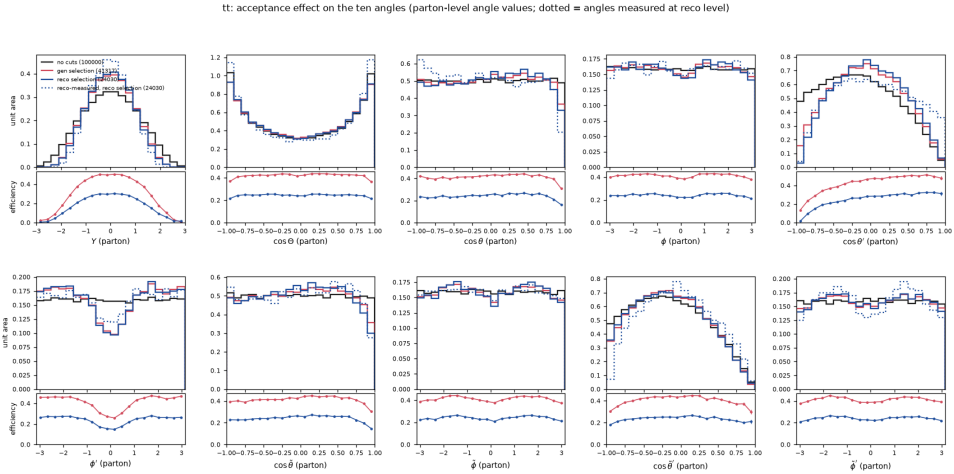
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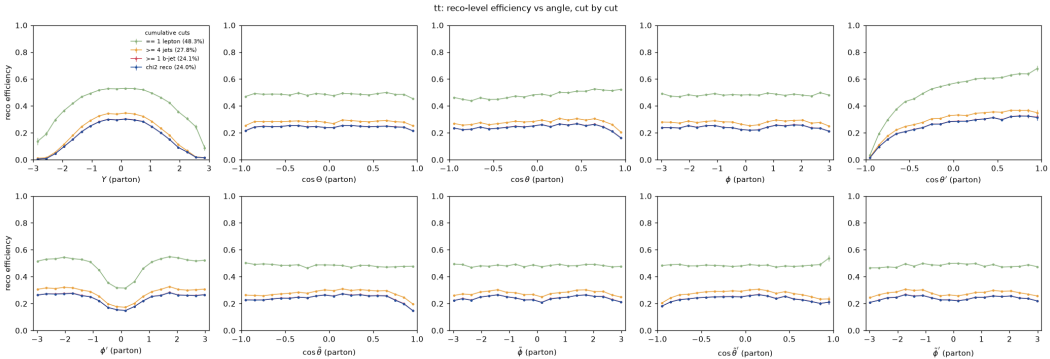
Acceptance: how the selection sculpts the angles

- Same (parton-level) angle values for every curve, so shape changes are acceptance only; dotted = angles measured at reco level (adds resolution and jet assignment)
- $\cos\theta' \to -1$, $\phi' \approx 0$: the lepton is emitted against the top's flight direction and fails $p_T > 25$ GeV (already at gen level, so not isolation)
- $|Y| \gtrsim 2$: $|\eta| < 2.5$ acceptance. $\cos\theta, \cos\tilde{\theta} \to +1$: the b quark goes backward and is too soft for the 25 GeV jet cut
- $\cos\Theta$ and the ϕ angles: nearly flat efficiency
- hh: the $t\bar{t}$-hypothesis χ^2 turns flat parton $\cos\Theta$ into $t\bar{t}$-like peaks at ± 1 (dotted): the reconstruction, more than the acceptance, makes hh look like $t\bar{t}$

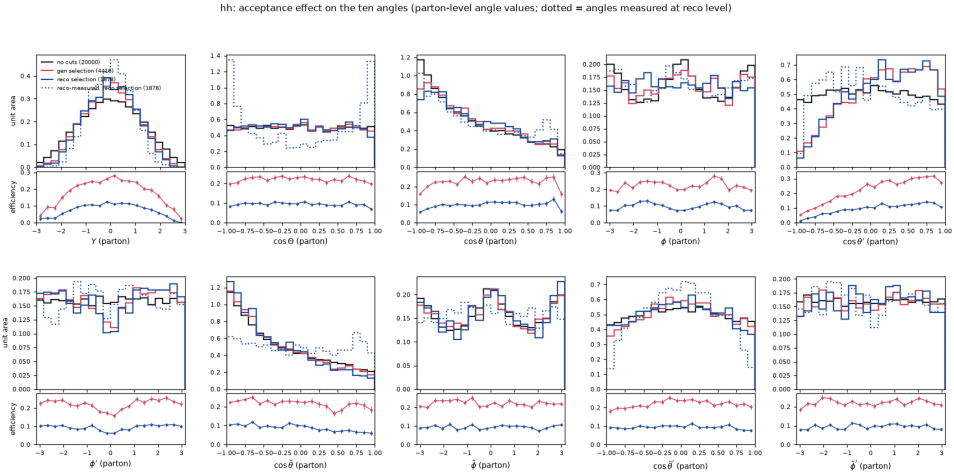
Acceptance, $t\bar{t}$: no cuts vs after cuts



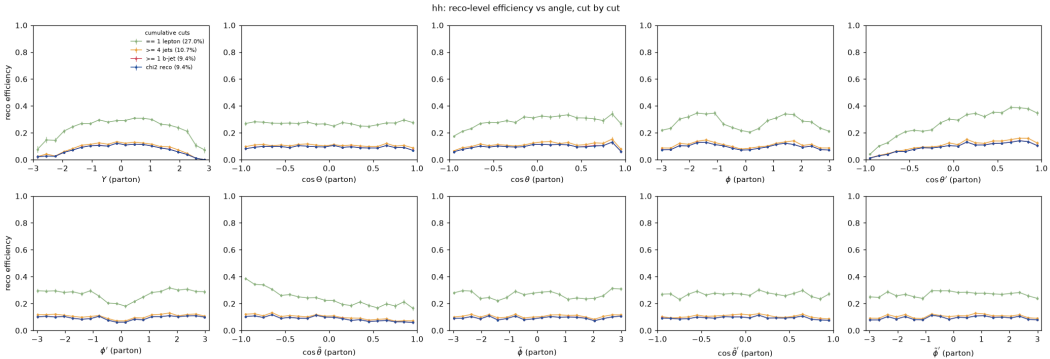
Acceptance, $t\bar{t}$: efficiency vs angle, cut by cut



Acceptance, hh: no cuts vs after cuts



Acceptance, hh: efficiency vs angle, cut by cut



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Summary (1/2): samples and angles

- $t\bar{t}$ samples (parton, gen, reco; ATLAS card, no pileup) produced and validated; hh (HEFT) likewise
- Ten angles as in the paper (diagram); parton level reproduces the SM W helicity fractions ($F_0 = 0.715 \pm 0.013$)
- Reco angles track parton angles when the jets are assigned correctly; combinatorics dominate the smearing
- ν p_z: 54 / 72 GeV (gen / reco, 68% half-width); the top-mass choice of the solution is not better than smaller $|p_z|$ (63% vs 65% closer to truth)

Summary (2/2): $f_{t\bar{t}}$ and hh

- $f_{t\bar{t}}$ from 14400 OSDE coefficients; AUC(hh vs independent $t\bar{t}$): parton 0.65, gen 0.53, reco 0.51
- Separation is lost at gen/reco because the $t\bar{t}$-hypothesis χ^2 makes hh look like $t\bar{t}$: e.g. a hadronic W^* (true median 32 GeV) is reconstructed near m_W; acceptance sculpts $\cos\theta'$, ϕ', Y
- Fit-sample vs independent-sample differences are entirely the self-inclusion bias (removed by leave-one-out)
- Only ∼50–350 coefficients matter: CV keeps 326, 459, 342 (parton, gen, reco); simplified f has fewer negative values and the same AUC; parton level resolves $t\bar{t}$ spin correlations
- Next: equiprobable-contour binning; truncated f as default?; Hermite scale; 8-angle set; an f_{hh} for A–B testing; pileup